

PHYSICS STUDY LAB

PRACTICE

Physics Solved

50 exam problems, worked step by step



GCSE

A-LEVEL

50 PROBLEMS

STEP BY STEP

Physics Study Lab

CONTENTS

Every problem is anchored to a chapter of the Physics Study Lab kit, and every solution tells you which page of it the theory lives on.

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HOW TO USE THIS BOOK

There is a right way and a wrong way, and the difference is about five minutes.

Every problem gets two pages. The question is on the left, on its own, with space to work. The full solution is on the right, where you cannot see it until you turn over.

That is deliberate. Reading a worked solution feels like learning and mostly is not. You follow each line, every line makes sense, and nothing sticks, because recognising a step someone else has taken is a far easier task than choosing it yourself. The learning happens in the five minutes before you turn the page – including, and especially, the minutes where you are stuck.

So: five minutes, pencil down, before you look. Get something on the page even if it is wrong. A wrong attempt you then correct is worth more than a right answer you only watched.

When you do turn over, read the trap first. Every solution carries a block marked THE TRAP: the specific wrong answer most students give, and the reason the brain produces it. If that was your answer, you have just learned more from this page than from getting it right.

Then check it makes sense. The last block of every solution tests the answer against something you already know – a number from the Highway Code, the time a kettle takes, the density of steel. That habit catches more errors in a real exam than any amount of re-reading.

If a step does not click, the footer of every solution tells you where the theory lives in the Physics Study Lab kit. This book does not explain physics. It shows you what to do with it.

About the level tags. Each problem is marked GCSE, A-LEVEL or BOTH. Boards differ slightly, so treat the tag as a guide rather than a ruling; where two syllabuses take different routes to the same answer, the solution shows both.

HOW TO READ A PHYSICS QUESTION

More marks are lost to the command word than to the maths.

State, Give, Name

Recall it. No working, no reason, no sentence. If you find yourself explaining, you are spending time you will not be paid for.

Define

Give the precise meaning, usually in the form of a word equation or a sentence that would stand up in a glossary.

Describe

Say WHAT happens. No reasons. A description that drifts into because has started answering a different question.

Explain

Say WHY it happens. Your answer needs the word because, or a word doing the same job. Marks here are for the link, not the fact.

Calculate

Produce a number. Show the equation, the substitution and the answer, with a unit. Working earns marks even when the final number is wrong.

Determine

Also a number, but you must first get the data yourself, from a graph, a table or a diagram. The hunt for the data is part of the question.

Show that

The answer is already printed in the question. Work to one more significant figure than the one quoted, then write a line stating that it agrees.

Estimate

Calculate from assumptions you state. The right order of magnitude scores. State every assumption you make, because they are marked too.

Suggest

Apply physics to a situation you have never seen. There is rarely one correct answer, and the examiner is looking for sound reasoning.

Compare

Both things, in every sentence. Use whereas and while. Two separate paragraphs describing one thing each is not a comparison.

Sketch

The shape of a graph, with the key values and intercepts labelled. Graph paper is not needed and accuracy of plotting is not marked.

Justify, Evaluate

State a conclusion and the evidence for it. Evaluate also expects the weaknesses, not only the case in favour.

SEVEN STEPS THAT SOLVE ANYTHING

Every solution in this book follows the same seven moves, in the same order.

1 Read it twice

The second time with a pencil, underlining every number and every command word. Most lost marks are lost here, before any physics happens.

2 Write down what you know

With units, in a column. The act of listing them is what reveals which equation is available to you.

3 Write down what you want

One line. If you cannot state what you are looking for, no equation will help you find it.

4 Draw it

A diagram, however rough, with the forces, rays or circuit on it. Problems 9, 10 and 42 in this book are all won or lost on the drawing.

5 Choose the equation that fits what you have

Not the one you remember best. The right equation contains what you were given and what you want, and nothing else.

6 Convert to SI before you substitute

Centimetres, minutes, megahertz and microcoulombs have ended more questions than any piece of physics. Convert first, every time.

7 Check three things

The unit, the size, and whether you answered the question that was asked. Half the traps in this book are caught by the second one alone.

PROBLEM

A motorway speed limit, in physics units

The UK motorway speed limit is 70 mph. Every physics equation you will use needs speed in metres per second. Convert it. Take 1 mile = 1609 m.

WHAT YOU KNOW

speed = 70 mph · 1 mile = 1609 m · 1 hour =
3600 s

WHAT YOU WANT

the same speed in m/s

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A motorway speed limit, in physics units

1 Turn the miles into metres

In one hour the car covers 70 miles.

$$70 \times 1609 \text{ m} = 112\,630 \text{ m}$$

2 Turn the hour into seconds

A speed is metres divided by seconds, so the hour has to go.

$$112\,630 \text{ m} \div 3600 \text{ s} = 31.29 \text{ m/s}$$

3 Quote it sensibly

The speed limit was given to 2 s.f., so 3 s.f. is as far as the data honestly reaches.

$$v = 31.3 \text{ m/s}$$

ANSWER

31.3 m/s

70 mph is just over 31 metres every second.

⚠ THE TRAP – most students answer 19.4 m/s

Dividing by 3.6. That converts km/h to m/s, not mph. Applied to miles per hour it throws away 38% of the speed. The “÷ 3.6” shortcut only becomes valid once the number is already in km/h.

DOES IT MAKE SENSE?

70 mph is about 112.6 km/h, and $112.6 \div 3.6 = 31.29 \text{ m/s}$. Two different routes, the same answer.

PROBLEM

Density of an object you cannot measure with a ruler

A stainless steel nut has a mass of 56.0 g. Dropped into a measuring cylinder, the water level rises from 50.0 cm³ to 57.0 cm³. Find its density in kg/m³.

WHAT YOU KNOW

mass = 56.0 g · water before = 50.0 cm³ · water
after = 57.0 cm³

WHAT YOU WANT

the density in kg/m³

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Density of an object you cannot measure with a ruler

1 The volume is the rise, not the final reading

The nut pushes aside its own volume of water, so the volume you want is the difference.

$$V = 57.0 - 50.0 = 7.0 \text{ cm}^3$$

2 Density = mass ÷ volume

Work in the units you were given, then convert once, at the end.

$$\rho = 56.0 \div 7.0 = 8.0 \text{ g/cm}^3$$

3 Convert to SI

1 g = 1/1000 kg and 1 cm³ = 1/1 000 000 m³, so the two conversions do not cancel: they leave a factor of 1000 upwards.

$$\rho = 8.0 \times 1000 = 8000 \text{ kg/m}^3$$

ANSWER

8000 kg/m³

8.0 g/cm³, which is 8000 kg/m³.

⚠ THE TRAP – most students answer 8.0 kg/m³

Keeping the g/cm³ number and simply changing the label. The answer is then out by a factor of 1000 – and it should look absurd on sight, because 8.0 kg/m³ is lighter than air.

DOES IT MAKE SENSE?

Stainless steel runs about 7900-8100 kg/m³. 8000 sits inside that range, so the answer describes a material that actually exists. That is the fastest check there is on a density.

PROBLEM

Two tugs, one barge

Two tugs pull a barge. One pulls with 8.0 kN due north, the other with 6.0 kN due east. Find the magnitude and direction of the resultant force.

WHAT YOU KNOW

$F_{\text{north}} = 8.0 \text{ kN}$ · $F_{\text{east}} = 6.0 \text{ kN}$ · angle
between them = 90°

WHAT YOU WANT

the resultant force: magnitude and direction

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Two tugs, one barge

1 Draw the two forces tip to tail

They are at 90° to each other, so the two forces and the resultant form a right-angled triangle.

the resultant is the hypotenuse

2 Pythagoras gives the magnitude

This is the whole point of calling force a vector: you cannot simply add the numbers.

$$F = \sqrt{(8.0^2 + 6.0^2)} = \sqrt{100} = 10.0 \text{ kN}$$

3 Trigonometry gives the direction

Measure it from north, the way a bearing is always quoted.

$$\theta = \tan^{-1}(6.0 \div 8.0) = 36.9^\circ \text{ east of north}$$

ANSWER

10.0 kN at 36.9° east of north

It is a 3-4-5 triangle scaled up by two, which is why the numbers come out clean.

⚠ THE TRAP – most students answer 14 kN

Adding the magnitudes. That is only correct when two forces point the same way. Here it overstates the pull by 40% – and it produces no direction at all, which is the clue that the method itself is wrong.

DOES IT MAKE SENSE?

A resultant must always lie between the difference of the two forces (2.0 kN) and their sum (14.0 kN). 10.0 kN sits inside that window.

PROBLEM

Reading a speed off a distance-time graph

A distance-time graph is a straight line. At $t = 2.0$ s the distance is 14.0 m; at $t = 6.0$ s it is 34.0 m. Find the speed.

WHAT YOU KNOW

point A = (2.0 s, 14.0 m) · point B = (6.0 s, 34.0 m)

WHAT YOU WANT

the speed of the object

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Reading a speed off a distance-time graph

1 Speed is the gradient, not the height

On a distance-time graph the gradient is distance per unit time, which is exactly what speed means.

$$v = \Delta d \div \Delta t$$

2 Take the differences

Use two points that are far apart; it keeps any reading error small.

$$\Delta d = 34.0 - 14.0 = 20.0 \text{ m and } \Delta t = 6.0 - 2.0 = 4.0 \text{ s}$$

3 Divide

$$v = 20.0 \div 4.0 = 5.0 \text{ m/s}$$

ANSWER

5.0 m/s

Constant speed, because the line is straight.

⚠ THE TRAP – most students answer 5.67 m/s

Taking a single point and dividing distance by time ($34.0 \div 6.0$). That only works if the line passes through the origin. This one does not – it starts at 4.0 m – so the single-point method comes out 13% too high.

DOES IT MAKE SENSE?

Test it on the other point: the line predicts $4.0 + 5.0 \times 2.0 = 14.0 \text{ m}$ at $t = 2.0 \text{ s}$, which is what the graph shows.

PROBLEM

Where the Highway Code braking distance comes from

A car travelling at 70 mph brakes hard, decelerating at 6.53 m/s^2 . How far does it travel between the brakes biting and the car stopping?

WHAT YOU KNOW

$u = 70 \text{ mph}$ (see problem 1) · $v = 0 \text{ m/s}$ · $a = -6.53 \text{ m/s}^2$

WHAT YOU WANT

the braking distance s

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Where the Highway Code braking distance comes from

1 Convert before you do anything else

Every SUVAT equation is written in SI units, and mph is not one.

$$u = 31.29 \text{ m/s}$$

2 Choose the equation without t in it

You are not asked for the time and you are not given it, so pick the equation that never mentions it.

$$v^2 = u^2 + 2as$$

3 The car stops, so $v = 0$

This is where most marks are lost: zero is the FINAL speed. The 70 mph is u , not v .

$$0 = 31.29^2 + 2(-6.53)s$$

4 Rearrange, and mind the two minus signs

Deceleration makes a negative, and it cancels against the minus from moving u^2 across. A positive distance is the sign that you did it right.

$$s = -979.1 \div (2 \times -6.53) = 979.1 \div 13.06 = 75.0 \text{ m}$$

ANSWER

75.0 m

About 75 metres – and that is only after the driver has already reacted.

⚠ THE TRAP – most students answer 375 m

Putting 70 straight into SUVAT. Because the equation squares the speed, a unit slip does not merely scale the answer, it scales it by the square: the result comes out 5.0 times too large.

DOES IT MAKE SENSE?

The Highway Code quotes a 75 m braking distance at 70 mph. This calculation reproduces it, which is the real lesson: the number in the booklet is physics, not a guess.

PROBLEM

A stone dropped from a cliff

A stone is dropped from a cliff 45 m above the sea. Ignoring air resistance, how long does it take to reach the water? Take $g = 9.81 \text{ m/s}^2$.

WHAT YOU KNOW

$$s = 45.0 \text{ m} \cdot u = 0 \text{ m/s} \cdot a = 9.81 \text{ m/s}^2$$

WHAT YOU WANT

the time of fall t

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A stone dropped from a cliff

1 "Dropped" means $u = 0$

That one word is worth a mark. "Thrown" would not be.

$$u = 0 \text{ m/s}$$

2 Choose the equation linking s , u , a and t

$$s = ut + \frac{1}{2}at^2, \text{ and with } u = 0 \text{ this becomes } s = \frac{1}{2}at^2$$

3 Rearrange for t

$$t = \sqrt{(2s \div a)} = \sqrt{(2 \times 45.0 \div 9.81)} = \sqrt{9.174}$$

4 Solve

$$t = 3.03 \text{ s}$$

ANSWER

3.03 s

It lands after about three seconds, hitting the water at 29.7 m/s.

⚠ THE TRAP – most students answer 4.59 s

Dividing the height by g . That treats 9.81 as a speed, when it is an acceleration – metres per second, per second. Dividing a distance by it does not produce a time at all; the units come out wrong, which is precisely how you catch this one before the examiner does.

DOES IT MAKE SENSE?

The impact speed works out at 29.7 m/s, about 107 km/h. Fast, but entirely believable for a 45 m drop – roughly a fifteen-storey building.

ANOTHER WAY IN

GCSE route. The equation $s = ut + \frac{1}{2}at^2$ is not on the GCSE sheet, but this problem is still within reach. Use $v^2 - u^2 = 2as$ to get the landing speed, $v = \sqrt{(2 \times 9.81 \times 45.0)} = 29.7 \text{ m/s}$, then $a = \Delta v \div \Delta t$ to get the time, $t = 29.7 \div 9.81 = 3.03 \text{ s}$. Same answer, two different syllabuses.

PROBLEM

The same cliff, but this time you run off it

From the same 45 m cliff, a stone is thrown horizontally at 12 m/s. How long is it in the air, and how far from the foot of the cliff does it land?

WHAT YOU KNOW

height = 45.0 m · horizontal speed = 12.0 m/s ·
 $g = 9.81 \text{ m/s}^2$

WHAT YOU WANT

the time of flight and the horizontal range

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The same cliff, but this time you run off it

1 Split the motion into two problems that do not talk to each other

Gravity acts vertically. It has no horizontal component, so it cannot change the horizontal speed – and the horizontal speed cannot change the fall.

$$\text{vertical: } u = 0, a = 9.81 \text{ m/s}^2 \mid \text{horizontal: } v = 12.0 \text{ m/s}, a = 0$$

2 The vertical half is problem 6, word for word

Same height, same g , same $u = 0$, so the same time.

$$t = \sqrt{(2 \times 45.0 \div 9.81)} = 3.029 \text{ s}$$

3 The horizontal half has no acceleration

Constant speed, so distance is just speed multiplied by time.

$$\text{range} = 12.0 \times 3.029 = 36.3 \text{ m}$$

ANSWER

$t = 3.03 \text{ s}$ and range = 36.3 m

Thrown or dropped, the stone hits the water at the same instant.

⚠ THE TRAP – most students answer a time longer than the 3.03 s of problem 6

Assuming that more horizontal speed keeps the stone up for longer. It does not. The vertical motion knows nothing about the horizontal motion. Drop one stone and fire another sideways at the same moment from the same height, and they land together – the fast one simply lands further out.

DOES IT MAKE SENSE?

Set the horizontal speed to zero and the problem should collapse back into problem 6. It does: $t = 3.03 \text{ s}$ either way.

PROBLEM

Why you feel heavy when a lift sets off

A 65 kg person stands on bathroom scales inside a lift. The lift accelerates upwards at 1.8 m/s^2 . What reading, in newtons, do the scales show?

WHAT YOU KNOW

$m = 65.0 \text{ kg}$ · $a = 1.8 \text{ m/s}^2$ upwards · $g = 9.81 \text{ m/s}^2$

WHAT YOU WANT

the normal contact force N from the scales

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Why you feel heavy when a lift sets off

1 The scales read the contact force, not the weight

A set of scales measures how hard you press on it, which by Newton's third law is how hard it presses back on you.

$$\text{reading} = N$$

2 Only two vertical forces act on the person

Weight down, normal force up. The person is accelerating upwards, so the upward force has to be the larger one.

$$W = mg = 65.0 \times 9.81 = 637.6 \text{ N, downwards}$$

3 Apply Newton's second law, taking up as positive

$$N - mg = ma \rightarrow N = m(g + a)$$

4 Substitute

$$N = 65.0 \times (9.81 + 1.8) = 65.0 \times 11.61 = 754.6 \text{ N}$$

ANSWER

755 N

Standing still the scales would read 638 N, so you appear 18% heavier.

⚠ THE TRAP – most students answer 117 N

Writing $N = ma$ and stopping there. That is the NET force, not the reading – the scales still have to hold the weight up as well. Leaving mg out gives an answer that is only 16% of the truth, which would mean the scales read LESS while accelerating upwards. That is the opposite of what anyone has ever felt in a lift.

DOES IT MAKE SENSE?

Divide the reading by g to see it as a mass: 76.9 kg. The scales behave as though the person had gained about 12 kg, which is exactly the heavy feeling in the first second of the ride.

PROBLEM

A crate on a ramp: does it slide?

A 4.0 kg crate sits on a ramp inclined at 25° . The coefficient of friction between crate and ramp is 0.30. Show whether the crate slides, and if it does, find its acceleration.

WHAT YOU KNOW

$m = 4.0 \text{ kg}$ · incline = 25° · $\mu = 0.30$ · $g = 9.81 \text{ m/s}^2$

WHAT YOU WANT

whether the crate slides, and its acceleration if it does

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A crate on a ramp: does it slide?

1 Resolve the weight along the slope and into it

Not horizontally and vertically. On an incline it is the slope that sets the useful pair of directions.

$$\text{along: } mg \sin 25^\circ = 16.58 \text{ N into: } mg \cos 25^\circ = 35.56 \text{ N}$$

2 The normal force balances only the part pressing in

The ramp pushes back exactly as hard as it is pressed, and only at right angles to its surface.

$$N = mg \cos 25^\circ = 35.56 \text{ N}$$

3 Find the largest friction force available

$$F(\text{max}) = \mu N = 0.30 \times 35.56 = 10.67 \text{ N}$$

4 Compare the two

The pull down the slope beats the most that friction can offer, so the crate moves.

$$16.58 \text{ N} > 10.67 \text{ N} \rightarrow \text{it slides}$$

5 Newton's second law, along the slope

$$a = (16.58 - 10.67) \div 4.0 = 1.48 \text{ m/s}^2$$

ANSWER

it slides, at 1.48 m/s^2

Down the slope, and only about 15% of free-fall acceleration.

△ THE TRAP – most students answer $F(\text{max}) = 11.8 \text{ N}$

Using $N = mg = 39.2 \text{ N}$. On a slope the surface never carries the whole weight, only the component pressing into it, so this overstates friction by 10%. The dangerous version of this mistake is on a shallow ramp: between about 16.7° and 17.5° the wrong method says the crate stays put when in fact it is already sliding.

DOES IT MAKE SENSE?

A crate slides whenever $\tan(\text{angle}) > \mu$. Here $\tan 25^\circ = 0.466$, which beats 0.30 – and notice the mass never appeared in that comparison at all.

PROBLEM

The same crate on a gentler ramp – so how big is the friction now?

The same 4.0 kg crate is placed on a ramp at 15° , where the coefficient of static friction is 0.40. The crate does not move. State the size of the friction force acting on it.

WHAT YOU KNOW

$m = 4.0 \text{ kg}$ · incline = 15° · $\mu_s = 0.40$ · $g = 9.81 \text{ m/s}^2$

WHAT YOU WANT

the actual friction force on the crate

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The same crate on a gentler ramp – so how big is the friction now?

1 First check that it really does stay put

$\tan 15^\circ = 0.268$, which is less than $\mu = 0.40 \rightarrow$ it does not move

2 Not moving means in equilibrium

Nothing is accelerating, so along the slope the forces must cancel exactly.

friction = the component of weight down the slope

3 Resolve the weight

$$mg \sin 15^\circ = 4.0 \times 9.81 \times 0.2588 = 10.2 \text{ N}$$

4 And that is the friction

Friction supplies exactly what is needed to hold the crate, and not a newton more.

$F = 10.2 \text{ N}$, up the slope

ANSWER

10.2 N

Up the slope, exactly balancing the pull of gravity along it.

⚠ THE TRAP – most students answer 15.2 N

Working out μN and quoting it. But μN is the MAXIMUM static friction, not the actual one: the law is $F \leq \mu N$, not $F = \mu N$. The crate needs only 10.2 N to stay still, so that is all friction supplies; the remaining 5.00 N is held in reserve. Quoting 15.2 N would leave a net force UP the slope, with the crate climbing it unaided.

DOES IT MAKE SENSE?

The size of the reserve is what tells you the crate is stable, and comfortably so: friction is using only 67% of what it has available.

PROBLEM

Dragging a suitcase through an airport

A suitcase is pulled 20 m across level floor by a handle held at 30° above the horizontal, with a force of 45 N along the handle. How much work does the pulling force do?

WHAT YOU KNOW

$F = 45 \text{ N}$ · $d = 20 \text{ m}$ · angle to the floor = 30°

WHAT YOU WANT

the work done by the pulling force

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Dragging a suitcase through an airport

1 Only the part of the force along the motion does work

The suitcase travels horizontally. The upward part of the pull moves it nowhere, so it does no work at all.

$$W = Fd \cos \theta$$

2 Resolve the force onto the direction of travel

$$F(\text{horizontal}) = 45 \cos 30^\circ = 45 \times 0.8660 = 38.97 \text{ N}$$

3 Multiply by the distance moved

$$W = 38.97 \times 20 = 779.4 \text{ J}$$

ANSWER

779 J

About 780 joules over the twenty metres.

⚠ THE TRAP – most students answer 900 J

Multiplying 45×20 and forgetting $\cos 30^\circ$, which assumes you are pulling horizontally. The vertical component of the pull is perfectly real – it is lifting weight off the wheels – but since the case never moves upwards, that component does zero work. Dropping the cosine inflates the answer by 15%.

DOES IT MAKE SENSE?

$\cos 30^\circ$ is 0.8660, so the answer has to come out at roughly 87% of 900 J. 779 J does.

PROBLEM

The slide, and why your weight makes no difference

A child starts from rest at the top of a smooth slide, 3.2 m above the ground. How fast are they moving at the bottom? The mass of the child is not given.

WHAT YOU KNOW

$h = 3.2 \text{ m}$ · $u = 0 \text{ m/s}$ · friction = none

WHAT YOU WANT

the speed at the bottom of the slide

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The slide, and why your weight makes no difference

1 Energy in equals energy out

The slide is smooth, so nothing is lost to friction: every joule of gravitational potential energy becomes kinetic energy.

$$mgh = \frac{1}{2}mv^2$$

2 The mass cancels

Which is why the question is allowed to withhold it – and why withholding it is the test.

$$gh = \frac{1}{2}v^2$$

3 Rearrange

$$v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 3.2} = \sqrt{62.78}$$

4 Solve

$$v = 7.92 \text{ m/s}$$

ANSWER

7.92 m/s

Independent of mass: a heavier child arrives at exactly the same speed.

⚠ THE TRAP – most students answer “not enough information, the mass is missing”

The mass is missing because it cancels. Every energy-conversion problem of this shape loses it on the second line. When a quantity is absent from a question, the first thing to test is whether the algebra removes it – not whether the examiner has made a mistake.

DOES IT MAKE SENSE?

That is 28.5 km/h, which is quick for a playground – and it should be, because the question deleted friction. A real slide of that height delivers noticeably less.

PROBLEM

Running up the stairs

A 58 kg student runs up a flight of stairs of total height 4.2 m in 6.5 s. Calculate the useful power they develop.

WHAT YOU KNOW

$m = 58 \text{ kg}$ · $h = 4.2 \text{ m}$ · $t = 6.5 \text{ s}$

WHAT YOU WANT

the useful power developed

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Running up the stairs

1 Find the work done against gravity

Only the height counts. Walking along each tread does no work against gravity, because gravity has no horizontal component.

$$W = mgh = 58 \times 9.81 \times 4.2 = 2390 \text{ J}$$

2 Power is work per second

$$P = W \div t$$

3 Divide

$$P = 2390 \div 6.5 = 367.7 \text{ W}$$

ANSWER

368 W

About 0.37 kW, or roughly half a horsepower.

⚠ THE TRAP – most students answer 2390 W

Quoting the work done and labelling it as power. Joules and watts are not interchangeable – a watt is a joule per second. The giveaway is that the time would then be left unused: if a question gives you a number and your answer never touches it, something has gone wrong.

DOES IT MAKE SENSE?

A fit person holds 150-250 W for a long time and several hundred watts in a short burst. 368 W sustained for 6.5 s is a hard sprint up the stairs, which is exactly what the question describes.

PROBLEM

Two trolleys that stick together

A 0.80 kg trolley moving at 3.0 m/s collides with a stationary 1.20 kg trolley. They stick together. Find their common velocity afterwards, and how much kinetic energy is lost.

WHAT YOU KNOW

$m_A = 0.80 \text{ kg}$ · $u_A = 3.0 \text{ m/s}$ · $m_B = 1.20 \text{ kg}$ · $u_B = 0 \text{ m/s}$

WHAT YOU WANT

the velocity after the collision, and the kinetic energy lost

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Two trolleys that stick together

1 Momentum is conserved in every collision, without exception

No external horizontal force acts on the pair, so the total momentum before equals the total after.

$$p(\text{before}) = 0.80 \times 3.0 + 1.20 \times 0 = 2.40 \text{ kg m/s}$$

2 Afterwards they move as a single object

$$p(\text{after}) = (0.80 + 1.20)v = 2.00v$$

3 Set them equal and solve

$$v = 2.40 \div 2.00 = 1.20 \text{ m/s}$$

4 Now compare the kinetic energies

Kinetic energy is not conserved here, and the second half of the question exists to make you prove it.

$$KE(\text{before}) = \frac{1}{2} \times 0.80 \times 3.0^2 = 3.60 \text{ J}$$

5 And afterwards

$$KE(\text{after}) = \frac{1}{2} \times 2.00 \times 1.20^2 = 1.44 \text{ J}$$

6 The difference is what was lost

It went into deforming the buffers, into sound, and into heat.

$$\text{lost} = 3.60 - 1.44 = 2.16 \text{ J}$$

ANSWER

$v = 1.20 \text{ m/s}$, and 2.16 J lost

60.0% of the kinetic energy disappears, while the momentum does not change at all.

△ THE TRAP – most students answer $v = 1.90 \text{ m/s}$

Conserving kinetic energy instead of momentum. Momentum is conserved in every collision; kinetic energy only in a perfectly elastic one, and two objects that stick together are the least elastic collision there is. Whenever a question says "stick", "couple" or "embed", kinetic energy is about to be lost.

DOES IT MAKE SENSE?

The same momentum is now carried by 2.50 times the mass, so the speed has to fall by that same factor: $3.0 \div 2.50 = 1.20 \text{ m/s}$.

PROBLEM

What an airbag actually does

A 70 kg driver travelling at 14 m/s is brought to rest in a crash. Against the steering column this takes 0.040 s; against an airbag it takes 0.25 s. Find the average force in each case.

WHAT YOU KNOW

$m = 70 \text{ kg}$ · $u = 14 \text{ m/s}$ · $v = 0 \text{ m/s}$ · t without
airbag = 0.040 s · t with airbag = 0.25 s

WHAT YOU WANT

the average force on the driver in each case

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

What an airbag actually does

1 Find the change in momentum

This is fixed by the crash itself, not by what the driver happens to hit.

$$\Delta p = m(v - u) = 70 \times (0 - 14) = -980 \text{ kg m/s}$$

2 Force is the rate of change of momentum

Newton's second law in its original form, and the more useful one here.

$$F = \Delta p \div \Delta t$$

3 Without the airbag

$$F = 980 \div 0.040 = 24\,500 \text{ N}$$

4 With the airbag

The identical momentum change, spread over six times as long.

$$F = 980 \div 0.25 = 3920 \text{ N}$$

ANSWER

24 500 N without the airbag, 3920 N with it

The airbag cuts the force by a factor of 6.25.

⚠ THE TRAP – most students answer “the airbag reduces the momentum”

It does not, and it could not. The driver goes from 14 m/s to rest either way, so Δp is 980 kg m/s in both cases. The airbag changes only Δt . That is the whole of its engineering: it is not soft so that it absorbs the blow, it is deep so that the blow takes longer.

DOES IT MAKE SENSE?

As multiples of body weight: 35.7g against the column and 5.71g against the airbag. A rollercoaster peaks near 5g. The airbag does not make the crash gentle – it moves it from a number the body cannot survive to one it can.

PROBLEM

How fast you can take a roundabout

A car drives round a flat roundabout of radius 25 m. The coefficient of friction between tyres and road is 0.70. Find the maximum speed before it skids. The mass of the car is not given.

WHAT YOU KNOW

$$r = 25 \text{ m} \cdot \mu = 0.70 \cdot g = 9.81 \text{ m/s}^2$$

WHAT YOU WANT

the maximum cornering speed

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

How fast you can take a roundabout

1 Ask what is providing the centripetal force

Nothing pushes the car round the bend except friction between tyre and road. On a flat roundabout there is no other horizontal force available at all.

$$\text{friction} = \text{centripetal force}$$

2 Write both sides out

$$\mu mg = mv^2 \div r$$

3 The mass cancels

It sits on both sides, which is exactly why the question is entitled to leave it out.

$$\mu g = v^2 \div r$$

4 Rearrange and solve

$$v = \sqrt{(\mu gr)} = \sqrt{(0.70 \times 9.81 \times 25)} = \sqrt{171.7} = 13.1 \text{ m/s}$$

ANSWER

13.1 m/s

That is 47.2 km/h, or 29.3 mph.

⚠ THE TRAP – most students answer “a heavier car has to go slower”

Mass cancels. A loaded lorry and an empty hatchback skid at the same speed on the same bend, because the extra mass needs more centripetal force but also presses the tyres down harder and earns precisely that much more friction. What does change the answer is μ – which is why the same roundabout is lethal in the wet.

DOES IT MAKE SENSE?

In the wet μ falls to roughly 0.40, giving 22.2 mph instead of 29.3 mph. That gap is the physics behind every “reduce speed in wet conditions” sign on the network.

PROBLEM

The pressure on a diver at the bottom of the pool

A diving pool is 3.8 m deep. Water has a density of 1000 kg/m^3 and atmospheric pressure at the surface is 101 000 Pa. Find the TOTAL pressure on a diver at the bottom.

WHAT YOU KNOW

$h = 3.8 \text{ m}$ · $\rho = 1000 \text{ kg/m}^3$ · atmospheric
pressure = 101 000 Pa · $g = 9.81 \text{ m/s}^2$

WHAT YOU WANT

the total pressure at the bottom of the pool

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The pressure on a diver at the bottom of the pool

1 Find the pressure the water alone adds

Pressure in a liquid depends only on depth, density and g . The shape of the pool makes no difference whatsoever.

$$P(\text{water}) = \rho gh = 1000 \times 9.81 \times 3.8 = 37\,278 \text{ Pa}$$

2 The atmosphere is still sitting on top of it

The water does not hold the air up. The whole column of atmosphere presses down on the pool surface and that pressure is passed on through the liquid.

$$P(\text{total}) = P(\text{atmospheric}) + P(\text{water})$$

3 Add them

$$P = 101\,000 + 37\,278 = 138\,278 \text{ Pa}$$

ANSWER

138 kPa

About 1.37 times atmospheric pressure, at a depth you could stand up in.

⚠ THE TRAP – most students answer 37 300 Pa

Giving only the water's contribution. That answers a different question – the GAUGE pressure, which is what a pressure gauge reads because it has air on its other side. The question asked for the total, and the atmosphere is 73.0% of it. Read whether a question wants total (absolute) or gauge pressure before you write anything down.

DOES IT MAKE SENSE?

Roughly 10 m of water is worth one atmosphere. At 3.8 m you would therefore expect about 0.37 extra atmospheres, and $37\,278 \div 101\,000 = 0.369$. The depth rule and the equation agree.

PROBLEM

Does it float, and does the size of it matter?

A rectangular pine block measures $20\text{ cm} \times 12\text{ cm} \times 5.0\text{ cm}$ and has a mass of 0.68 kg . Water has a density of 1000 kg/m^3 . Show whether the block floats, and state what fraction of it sits below the waterline.

WHAT YOU KNOW

dimensions = $0.20 \times 0.12 \times 0.050\text{ m}$ · mass = 0.68 kg · $\rho_{\text{water}} = 1000\text{ kg/m}^3$

WHAT YOU WANT

whether it floats, and the fraction submerged

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Does it float, and does the size of it matter?

1 Work in metres from the start

Mixing centimetres into an SI calculation is the single most common way to lose this mark.

$$V = 0.20 \times 0.12 \times 0.050 = 0.0012 \text{ m}^3$$

2 Find the density

$$\rho = 0.68 \div 0.0012 = 567 \text{ kg/m}^3$$

3 Compare with the liquid, not with anything else

Floating is decided by density against density. Mass on its own tells you nothing.

$$567 < 1000 \rightarrow \text{it floats}$$

4 A floating object sinks until it displaces its own weight

So the fraction submerged is just the ratio of the two densities.

$$\text{fraction} = 567 \div 1000 = 0.567, \text{ or } 56.7\%$$

ANSWER

it floats, with 56.7% of it submerged

Just over half the block sits below the waterline.

⚠ THE TRAP – most students answer “a bigger block of the same wood would sink”

It would not. Double every dimension and the mass goes up eight-fold – but so does the volume, so the density is unchanged and the block floats exactly as high in the water. Floating is a property of the material, not of the lump. A pine log and a pine matchstick float with the same fraction submerged.

DOES IT MAKE SENSE?

Pine runs about 350-600 kg/m³ dry. 567 sits inside that band, so the number describes real timber – and it predicts the block floating a little over half under, which is what a plank in a pond actually does.

PROBLEM

A trolley jack, and the thing it does not do

In a hydraulic jack the small piston has an area of 2.0 cm^2 and the large piston 150 cm^2 . A force of 90 N is applied to the small piston. Find the force on the large one, and the mass it could lift.

WHAT YOU KNOW

$A_{\text{small}} = 2.0 \text{ cm}^2 = 2.0 \times 10^{-4} \text{ m}^2$ · $A_{\text{large}} = 150 \text{ cm}^2 = 150 \times 10^{-4} \text{ m}^2$ · $F_{\text{small}} = 90 \text{ N}$

WHAT YOU WANT

the output force, and the mass it can lift

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A trolley jack, and the thing it does not do

1 Pascal: the pressure is the same everywhere in the fluid

This is the whole principle. The liquid cannot tell the two pistons apart; it only knows pressure.

$$P = F \div A = 90 \div 0.00020 = 450\,000 \text{ Pa}$$

2 Apply that same pressure to the larger area

$$F = PA = 450\,000 \times 0.0150 = 6750 \text{ N}$$

3 Turn the force into a mass

$$m = F \div g = 6750 \div 9.81 = 688 \text{ kg}$$

ANSWER

6750 N, enough to lift 688 kg

The jack multiplies the force by 75, which is exactly the ratio of the two areas.

⚠ THE TRAP – most students answer “so the jack multiplies the energy by 75 as well”

It does not, and nothing could. Energy is conserved: to raise the large piston by 1 cm you must push the small one down 75 cm, because the same volume of oil has to move. Multiply force by 75 and divide distance by 75 and the work done is identical at both ends. A hydraulic jack trades distance for force – it does not create anything.

DOES IT MAKE SENSE?

688 kg is comfortably more than one corner of a family car, which is precisely the job a trolley jack is built for. And the force ratio came out at 75, matching the area ratio exactly, as Pascal requires.

PROBLEM

Why a steel block weighs less underwater

A steel block of volume $2.5 \times 10^{-4} \text{ m}^3$ and density 7800 kg/m^3 hangs from a spring balance, fully submerged in water. What does the balance read?

WHAT YOU KNOW

$V = 2.5 \times 10^{-4} \text{ m}^3$ · $\rho_{\text{steel}} = 7800 \text{ kg/m}^3$ · $\rho_{\text{water}} = 1000 \text{ kg/m}^3$ · $g = 9.81 \text{ m/s}^2$

WHAT YOU WANT

the apparent weight shown on the balance

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Why a steel block weighs less underwater

1 Find the real weight first

$$W = \rho V g = 7800 \times 0.00025 \times 9.81 = 19.13 \text{ N}$$

2 Archimedes: upthrust equals the weight of fluid displaced

Displaced – so the density in this line is the WATER's, never the block's.

$$U = \rho(\text{water}) \times V \times g = 1000 \times 0.00025 \times 9.81 = 2.45 \text{ N}$$

3 The balance reads what is left

$$\text{reading} = 19.13 - 2.45 = 16.68 \text{ N}$$

ANSWER

16.7 N

The block appears to lose 2.45 N, about 12.8% of its weight.

⚠ THE TRAP – most students answer a reading of 0 N

Using the block's own density in the upthrust line instead of the water's. That makes the upthrust equal the weight for every object ever submerged, so nothing would ever sink. The moment an answer implies that steel floats, the error is in which density went into the formula.

DOES IT MAKE SENSE?

The fraction of weight lost should be $\rho(\text{water}) \div \rho(\text{steel}) = 1000 \div 7800 = 0.128$. The calculation gives 0.128. Two independent routes to the same figure.

PROBLEM

Water through a narrowing pipe

Water flows at 1.2 m/s through a pipe of internal diameter 50 mm, which narrows to 25 mm. Find the speed in the narrow section, and say whether the pressure there is higher or lower, and by how much.

WHAT YOU KNOW

$v_1 = 1.2 \text{ m/s}$ · $d_1 = 50 \text{ mm}$ · $d_2 = 25 \text{ mm}$ · $\rho = 1000 \text{ kg/m}^3$

WHAT YOU WANT

the speed in the narrow section and the pressure change

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Water through a narrowing pipe

1 The same water per second has to pass every cross-section

Nothing is being stored or lost along the pipe, so $A_1v_1 = A_2v_2$.

$$A_1v_1 = A_2v_2$$

2 Area goes with the SQUARE of the diameter

The diameter halves, so the area quarters – the step that decides the whole answer.

$$A_1 \div A_2 = (50 \div 25)^2 = 4$$

3 So the speed goes up four-fold

$$v_2 = 1.2 \times 4 = 4.8 \text{ m/s}$$

4 Bernoulli: faster flow means lower pressure

The water speeds up, so something must have pushed it. That push is a pressure DROP from wide section to narrow.

$$\Delta P = \frac{1}{2}\rho(v_2^2 - v_1^2) = 500 \times (23.04 - 1.44) = 10\,800 \text{ Pa lower}$$

ANSWER

4.8 m/s, at a pressure 10 800 Pa LOWER

The narrow section is the fast section, and the fast section is the low-pressure one.

⚠ THE TRAP – most students answer “the pressure is higher in the narrow bit, because the water is squeezed”

This is the most persistent wrong intuition in fluids, and it is backwards. Squeezing a static fluid does raise its pressure – but this fluid is not static. The water arrives at the constriction and SPEEDS UP, and the only thing that can accelerate it is a force, which means a push from high pressure behind towards low pressure ahead. The pressure has to fall for the water to go faster. It is why an aerofoil lifts and why two sheets of paper pull together when you blow between them.

DOES IT MAKE SENSE?

10 800 Pa is about 11% of atmospheric pressure, or the pressure under 1.1 m of water. Small, but easily enough to show on a manometer across the constriction – which is exactly how a Venturi flow meter measures flow rate.

PROBLEM

How long the kettle really takes

A kettle holds 1.5 kg of water at 18 °C. The specific heat capacity of water is 4180 J/kg°C and the kettle delivers 2.8 kW. How much energy is needed to bring it to 100 °C, and how long should that take?

WHAT YOU KNOW

$m = 1.5 \text{ kg}$ · $c = 4180 \text{ J/kg}^\circ\text{C}$ · start = 18 °C ·
finish = 100 °C · $P = 2800 \text{ W}$

WHAT YOU WANT

the energy required, and the time taken

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

How long the kettle really takes

1 ΔT is the CHANGE in temperature, not the final one

The water does not start at zero. This one line is the whole problem.

$$\Delta T = 100 - 18 = 82\text{ }^{\circ}\text{C}$$

2 Apply the specific heat capacity equation

$$Q = mc\Delta T = 1.5 \times 4180 \times 82 = 514\,140\text{ J}$$

3 Power is energy per second, so time is energy over power

$$t = Q \div P = 514\,140 \div 2800 = 184\text{ s}$$

ANSWER

514 kJ, taking 184 s

Just over three minutes – which is how long a kettle actually takes.

⚠ THE TRAP – most students answer 627 000 J

Using 100 as ΔT , because 100 is the number in the question. But the water begins at 18 °C, so only 82 degrees of heating are needed. Taking the final temperature instead of the rise overstates the energy by 22% – and it would predict a kettle that takes 224 s, which anyone who has made tea knows is wrong.

DOES IT MAKE SENSE?

A 3 kW kettle boils a litre and a half in roughly three minutes from cold tap water. The calculation gives 184 s. When a thermal answer can be checked against something in your own kitchen, check it.

PROBLEM

Melting ice without warming it at all

How much energy is needed to melt 0.25 kg of ice that is already at 0 °C, into water at 0 °C? The specific latent heat of fusion of water is 334 000 J/kg.

WHAT YOU KNOW

$m = 0.25 \text{ kg}$ · $L(f) = 334\,000 \text{ J/kg}$ · start = ice at 0 °C · finish = water at 0 °C

WHAT YOU WANT

the energy needed to melt it

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Melting ice without warming it at all

1 Notice that the temperature never changes

It goes in at 0 °C and comes out at 0 °C. So $mc\Delta T$ contributes nothing at all here – ΔT is zero.

$$\Delta T = 0, \text{ so } Q = mc\Delta T = 0$$

2 All the energy goes into breaking the bonds, not into speeding molecules up

Temperature measures the kinetic energy of the particles. During a change of state the energy goes into potential energy instead, pulling the lattice apart.

$$Q = mL(f)$$

3 Substitute

$$Q = 0.25 \times 334\,000 = 83\,500 \text{ J}$$

ANSWER

83.5 kJ

And not one degree of temperature rise to show for it.

⚠ THE TRAP – most students answer “the ice warms up as it melts”

It does not. An ice-water mixture sits at 0 °C and stays there until the last of the ice has gone, no matter how much energy you pour in. That is why a drink with ice in it holds its temperature and a drink without it does not – and it is why the flat section of a heating curve is flat.

DOES IT MAKE SENSE?

Put that 83.5 kJ into the resulting water instead and it would raise its temperature by $83\,500 \div (0.25 \times 4180) = 79.9$ °C. Merely melting ice costs as much energy as heating the same water from freezing to nearly boiling.

PROBLEM

Mixing hot and cold water

0.30 kg of water at 80 °C is mixed with 0.50 kg of water at 20 °C in an insulated container. Find the final temperature.

WHAT YOU KNOW

m hot = 0.30 kg at 80 °C · m cold = 0.50 kg at 20 °C · heat lost to surroundings = none

WHAT YOU WANT

the final temperature of the mixture

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Mixing hot and cold water

1 The energy the hot water loses, the cold water gains

Nothing escapes the container, so the two must balance exactly.

$$m(\text{hot}) c (80 - T) = m(\text{cold}) c (T - 20)$$

2 The specific heat capacity cancels

It is the same liquid on both sides, so its value never has to be looked up.

$$0.30(80 - T) = 0.50(T - 20)$$

3 Expand and collect

$$24 - 0.30T = 0.50T - 10 \rightarrow 34.0 = 0.80T$$

4 Solve

$$T = 34.0 \div 0.80 = 42.5 \text{ }^{\circ}\text{C}$$

ANSWER

42.5 °C

Below the midpoint, because there is more cold water than hot.

△ THE TRAP – most students answer 50 °C

Averaging the two temperatures. That is only correct when the two masses are equal, and here they are not: there is 1.67 times as much cold water as hot, so the mixture has to settle nearer the cold end. The average is a special case, not the rule.

DOES IT MAKE SENSE?

The answer must lie between 20 and 80 °C, and nearer 20 because the cold mass is larger. 42.5 °C satisfies both. Check it the other way: the hot water falls 37.5 °C and the cold rises 22.5 °C, and $0.30 \times 37.5 = 0.50 \times 22.5$. Energy balances.

PROBLEM

Why your tyre pressure is wrong after a motorway run

A car tyre is checked cold at 15 °C and holds air at an absolute pressure of 3.4 bar. After a long drive the air inside has reached 45 °C. The volume is effectively unchanged. Find the new pressure.

WHAT YOU KNOW

$P_1 = 3.4 \text{ bar absolute}$ · $T_1 = 15 \text{ °C}$ · $T_2 = 45 \text{ °C}$ ·
 $V = \text{constant}$

WHAT YOU WANT

the new absolute pressure

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Why your tyre pressure is wrong after a motorway run

1 Convert both temperatures to kelvin, before anything else

The gas laws are proportional relationships, and a proportion needs a scale that starts at true zero. Celsius does not; kelvin does.

$$T_1 = 15 + 273 = 288 \text{ K} \quad T_2 = 45 + 273 = 318 \text{ K}$$

2 Constant volume, so pressure is proportional to temperature

$$P_1 \div T_1 = P_2 \div T_2$$

3 Rearrange and substitute

$$P_2 = 3.4 \times 318 \div 288 = 3.75 \text{ bar}$$

ANSWER

3.75 bar absolute

A rise of 0.35 bar, which is why tyres are always checked cold.

⚠ THE TRAP – most students answer 10.2 bar

Using degrees Celsius, so that 45 °C looks like three times 15 °C and the pressure appears to triple. It does not: in kelvin the temperature rose from 288 to 318, a rise of only 10%. Celsius has an arbitrary zero, so ratios taken in it are meaningless. Any gas-law answer that changes by a factor of three for a warm afternoon is a units error.

DOES IT MAKE SENSE?

The absolute temperature went up by 10%, so at constant volume the pressure must go up by 10% too: $3.4 \times 1.104 = 3.75 \text{ bar}$. The percentage check and the equation agree, and both match the few tenths of a bar a garage gauge actually shows after a run.

PROBLEM

A bubble on its way to the surface

A bubble of volume 1.0 cm^3 forms 12 m down in a lake and rises to the surface. The temperature does not change. Atmospheric pressure is 101 000 Pa and the water's density is 1000 kg/m^3 . Find the bubble's volume at the surface.

WHAT YOU KNOW

$V_1 = 1.0 \text{ cm}^3$ · depth = 12 m · atmospheric pressure = 101 000 Pa · temperature = constant

WHAT YOU WANT

the volume of the bubble at the surface

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A bubble on its way to the surface

1 Find the ABSOLUTE pressure down at the bubble

The water above it and the atmosphere above that both press on it. Boyle's law needs the total, not just the water's share.

$$P_1 = 101\,000 + (1000 \times 9.81 \times 12) = 101\,000 + 117\,700 = 218\,700 \text{ Pa}$$

2 At the surface only the atmosphere is left

$$P_2 = 101\,000 \text{ Pa}$$

3 Temperature is constant, so Boyle's law applies

$$P_1 V_1 = P_2 V_2$$

4 Rearrange and solve

$$V_2 = 1.0 \times 218\,700 \div 101\,000 = 2.17 \text{ cm}^3$$

ANSWER

2.17 cm³

The bubble more than doubles on the way up, without a single extra molecule entering it.

⚠ THE TRAP – most students answer 1.17 cm³

Leaving the atmosphere out and using only the water's 117 700 Pa. Boyle's law is written in absolute pressures, and a bubble 12 m down is carrying the whole sky on its shoulders as well as the lake.

Dropping the atmosphere here makes the bubble SHRINK as it rises, which is the opposite of what anyone has ever seen.

DOES IT MAKE SENSE?

About every 10 m of water adds one atmosphere. At 12 m the bubble sits under roughly 2.2 atmospheres, so at the surface it should be about 2.2 times bigger – and 2.17 cm³ is 2.17 times 1.0. It is also the reason divers are told never to hold their breath on the way up.

PROBLEM

A power station that throws away more than it sells

A thermal power station burns fuel at a rate of 2400 MW and delivers 950 MW of electricity to the grid. Find its efficiency and the power it rejects as waste heat.

WHAT YOU KNOW

power in = 2400 MW · electrical power out = 950 MW

WHAT YOU WANT

the efficiency and the wasted power

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A power station that throws away more than it sells

1 Efficiency is useful output over total input

Both are powers here, so no conversion to energy is needed – the seconds cancel.

$$\text{efficiency} = 950 \div 2400 = 0.396$$

2 Express it as a percentage

$$= 39.6\%$$

3 Everything that is not useful output leaves as heat

Energy is conserved. It did not vanish; it went up the cooling towers and into the river.

$$\text{waste} = 2400 - 950 = 1450 \text{ MW}$$

ANSWER

39.6% efficient, rejecting 1450 MW

The station discards more than half as much again as it sells.

△ THE TRAP – most students answer “so 60.4% is being wasted through bad engineering”

Most of it is not an engineering failure at all – it is forbidden by the second law of thermodynamics. Any engine that turns heat into work must dump some heat at a lower temperature; that rejection is what makes the process possible, not a leak in it. Calling the whole of it waste misreads the physics.

DOES IT MAKE SENSE?

Real coal and gas thermal stations run at about 38-42%. 39.6% sits right inside that band, which is the sign the numbers in the question describe a real plant rather than an invented one.

ANOTHER WAY IN

A-level extension – how much of that loss is unavoidable. Carnot’s limit says the best possible efficiency is $1 - T(\text{cold}) \div T(\text{hot})$, using absolute temperatures. With steam at 550 °C (823 K) and cooling water at 20 °C (293 K): $1 - 293 \div 823 = 0.644$, or 64.4%. So the laws of physics cap this station at 64.4% and the engineers reached 39.6%. The gap worth criticising is that one – not the whole 60.4%.

PROBLEM

What one window costs you overnight

A window measures $1.4 \text{ m} \times 1.1 \text{ m}$ and loses heat at a rate given by $P = UA\Delta T$, where $U = 2.8 \text{ W/m}^2\text{°C}$. Inside it is 21 °C and outside 4 °C . Find the rate of heat loss, and the energy lost in 24 hours.

WHAT YOU KNOW

dimensions = $1.4 \times 1.1 \text{ m}$ · $U = 2.8 \text{ W/m}^2\text{°C}$ ·
inside = 21 °C · outside = 4 °C

WHAT YOU WANT

the rate of heat loss, and the energy lost in a day

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

What one window costs you overnight

1 Area, and the temperature DIFFERENCE

As always in thermal problems, what drives the flow is the difference, not either temperature on its own.

$$A = 1.4 \times 1.1 = 1.54 \text{ m}^2 \quad \Delta T = 21 - 4 = 17 \text{ }^\circ\text{C}$$

2 Substitute into the equation given

The answer is in watts, so it is a RATE – joules every second, continuously.

$$P = 2.8 \times 1.54 \times 17 = 73.3 \text{ W}$$

3 Energy is that rate multiplied by the time it runs for

24 hours is 86 400 seconds. Convert before multiplying, not after.

$$E = Pt = 73.30 \times 86\,400 = 6\,333\,120 \text{ J}$$

4 And in the unit the energy bill uses

$$E = 73.3 \text{ W} \times 24 \text{ h} = 1.76 \text{ kWh}$$

ANSWER

73.3 W, which is 6.33 MJ or 1.76 kWh per day

That one window leaks heat as steadily as a 73 W bulb left on, day and night.

△ THE TRAP – most students answer quoting 6 333 120 W, or 73.3 J

Swapping the two. A watt is a joule per second: the first answer is a rate and the second is a total, and they are not interchangeable labels for the same number. The giveaway is the time – a rate is true at every instant and never mentions a duration, while an energy is meaningless until you say over how long.

DOES IT MAKE SENSE?

1.76 kWh a day from one window. A house with ten similar windows loses around 17.6 kWh a day through the glazing alone, which is a serious share of a winter heating bill – and exactly why double glazing is sold on its U-value.

PROBLEM

The one equation everything else is built on

A torch bulb has a potential difference of 3.0 V across it and a current of 0.25 A through it. Find its resistance, and the power it dissipates.

WHAT YOU KNOW

$$V = 3.0 \text{ V} \cdot I = 0.25 \text{ A}$$

WHAT YOU WANT

the resistance of the bulb, and its power

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The one equation everything else is built on

1 Start from the definition, not from a memorised triangle

Resistance is how much potential difference it takes to drive one amp through something. That sentence IS the equation.

$$V = IR \rightarrow R = V \div I$$

2 Substitute

$$R = 3.0 \div 0.25 = 12.0 \, \Omega$$

3 Power is current times potential difference

Volts are joules per coulomb and amps are coulombs per second, so multiplying them gives joules per second, which is watts.

$$P = VI = 3.0 \times 0.25 = 0.750 \, \text{W}$$

ANSWER

12.0 Ω , dissipating 0.750 W

A small torch bulb, and three quarters of a watt of light and heat.

⚠ THE TRAP – most students answer 0.0833 Ω

Dividing the wrong way round: $I \div V$ instead of $V \div I$. The triangle is remembered upside down about as often as it is remembered correctly. The defence is not a better mnemonic, it is a sense of scale – 0.0833 Ω is the resistance of a short piece of copper wire, not of something that glows.

DOES IT MAKE SENSE?

Check it the other way: 12.0 Ω carrying 0.25 A needs $12.0 \times 0.25 = 3.00$ V across it, which is what the question said. And torch bulbs do sit in the range of a few ohms to a few tens of ohms.

PROBLEM

Two resistors in series

A $220\ \Omega$ resistor and a $330\ \Omega$ resistor are connected in series across a $12\ \text{V}$ supply. Find the current, and the potential difference across each resistor.

WHAT YOU KNOW

$R_1 = 220\ \Omega$ · $R_2 = 330\ \Omega$ · $V_{\text{supply}} = 12\ \text{V}$

WHAT YOU WANT

the current, and the potential difference across each resistor

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Two resistors in series

1 In series, resistances simply add

There is only one path, so the charge has to fight its way through both. The obstacles come one after the other.

$$R(\text{total}) = 220 + 330 = 550 \, \Omega$$

2 One path means ONE current, the same everywhere

Charge is not used up as it goes round. Whatever leaves the battery comes back to it.

$$I = V \div R = 12 \div 550 = 0.02182 \, \text{A}$$

3 Now find each potential difference separately

Each resistor gets the share of the voltage that its own resistance demands.

$$V_1 = 0.02182 \times 220 = 4.80 \, \text{V}$$

4 And the second

$$V_2 = 0.02182 \times 330 = 7.20 \, \text{V}$$

ANSWER

21.8 mA, with 4.80 V and 7.20 V across them

The larger resistor takes the larger share of the voltage, in exactly the same ratio as the resistances.

⚠ THE TRAP – most students answer 6 V across each

Splitting the supply equally. The voltage divides in the ratio of the resistances, not evenly: 330 is 1.50 times 220, so it takes 1.50 times the voltage. Equal sharing only happens when the resistors are equal, which is a special case rather than the rule.

DOES IT MAKE SENSE?

The two potential differences must add up to the supply: $4.80 + 7.20 = 12.00 \, \text{V}$. They do. If they do not add to the supply voltage, something is wrong, and this check costs five seconds.

PROBLEM

The same two resistors in parallel

The same $220\ \Omega$ and $330\ \Omega$ resistors are now connected in PARALLEL across the same 12 V supply. Find the total resistance and the total current.

WHAT YOU KNOW

$R_1 = 220\ \Omega$ · $R_2 = 330\ \Omega$ · $V_{\text{supply}} = 12\text{ V}$

WHAT YOU WANT

the total resistance and the current drawn from the supply

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The same two resistors in parallel

1 In parallel it is the CONDUCTANCES that add, not the resistances

You have not made the path harder, you have opened a second one. Two doors out of a room let more people through than one.

$$1 \div R = 1 \div 220 + 1 \div 330$$

2 Put it over a common denominator

$$1 \div R = 3 \div 660 + 2 \div 660 = 5 \div 660$$

3 Invert - and this is the step people forget

The equation gives you $1/R$. It does not give you R until you turn it over.

$$R = 660 \div 5 = 132 \, \Omega$$

4 Now the total current

$$I = 12 \div 132 = 0.0909 \, \text{A}$$

ANSWER

132 Ω , drawing 90.9 mA

Over four times the current the series version drew, from the same two components and the same battery.

⚠ THE TRAP – most students answer 550 Ω , giving 21.8 mA

Adding the resistances, exactly as in series. This is the single most common mistake in school electricity, and it is 4.17 times out. It also fails the test that costs nothing: a parallel combination must ALWAYS come out smaller than the smallest resistor in it. Adding gives 550 Ω , which is larger than both. If your parallel answer is bigger than any single resistor, you have added when you should have inverted.

DOES IT MAKE SENSE?

Two independent checks. First, 132 Ω is smaller than 220 Ω , as a parallel total must be. Second, add the branch currents: $12 \div 220 = 0.0546 \, \text{A}$ and $12 \div 330 = 0.0364 \, \text{A}$, and $0.0546 + 0.0364 = 0.0909 \, \text{A}$, matching the total. Both routes agree.

PROBLEM

A potential divider, and which resistor goes on top

A $10\text{ k}\Omega$ resistor and a $4.7\text{ k}\Omega$ resistor are connected in series across a 9.0 V supply. Find the potential difference across the $4.7\text{ k}\Omega$ resistor.

WHAT YOU KNOW

$R_1 = 10\text{ k}\Omega$ · $R_2 = 4.7\text{ k}\Omega$ · $V_{\text{supply}} = 9.0\text{ V}$

WHAT YOU WANT

the potential difference across the $4.7\text{ k}\Omega$ resistor

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A potential divider, and which resistor goes on top

1 The two resistors share the supply in proportion to their resistance

Same current through both, so $V = IR$ means the voltages are in the same ratio as the resistances.

$$V_2 = V \times R_2 \div (R_1 + R_2)$$

2 The resistor you are asked about goes on TOP of the fraction

This is the only thing that ever goes wrong with a potential divider. The denominator is always the total.

$$V_2 = 9.0 \times 4700 \div 14\,700$$

3 Solve

$$V_2 = 2.88 \text{ V}$$

ANSWER

2.88 V

Leaving 6.12 V across the 10 k Ω .

⚠ THE TRAP – most students answer 6.12 V

Putting the OTHER resistor on top: $9.0 \times 10\,000 \div 14\,700$. That is the voltage across the 10 k Ω , which is a perfectly good answer to a question nobody asked. Both numbers will appear on your calculator and both look plausible, which is precisely why this one is worth a check.

DOES IT MAKE SENSE?

The 4.7 k Ω is the SMALLER resistor, so it must take LESS than half the supply – less than 4.5 V. 2.88 V passes; 6.12 V would have failed instantly. And the two shares add back to the supply: $2.88 + 6.12 = 9.00$ V.

PROBLEM

Three equations for power, and picking the one you can actually use

A $56\ \Omega$ resistor carries a current of $0.35\ \text{A}$. Calculate the power it dissipates. The supply voltage is not given.

WHAT YOU KNOW

$$R = 56\ \Omega \cdot I = 0.35\ \text{A}$$

WHAT YOU WANT

the power dissipated in the resistor

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Three equations for power, and picking the one you can actually use

1 Write down all three versions before choosing

They are the same equation wearing different clothes, because $V = IR$ lets you swap between them.

$$P = VI \quad P = I^2R \quad P = V^2 \div R$$

2 Choose the one that uses only what you have

You were given I and R , so the middle form needs nothing else. The other two would require a voltage you have not been told.

$$P = I^2R$$

3 Substitute, and square the current before multiplying

$$P = 0.35^2 \times 56 = 0.1225 \times 56 = 6.86 \text{ W}$$

ANSWER

6.86 W

Nearly seven watts, in a component the size of a grain of rice.

⚠ THE TRAP – most students answer reaching for $P = VI$ and using the supply voltage

The V in every one of these equations is the potential difference across THAT component, never the voltage of the battery driving the circuit. In a circuit with anything else in it, those two are different numbers, and using the supply voltage silently credits one resistor with the power of the whole circuit.

DOES IT MAKE SENSE?

Work out the missing voltage and check with a different equation: $V = IR = 0.35 \times 56 = 19.6 \text{ V}$, so $P = VI = 19.6 \times 0.35 = 6.86 \text{ W}$. The two forms agree. Worth noting too that 6.86 W would destroy a standard 0.25 W resistor, which is why power ratings exist.

PROBLEM

What a tumble dryer really costs you in energy

A tumble dryer is rated at 2.2 kW and runs for 1.5 hours per cycle, four cycles a week. Find the energy used per cycle in kilowatt-hours and in joules, and the energy used in a year.

WHAT YOU KNOW

$P = 2.2 \text{ kW}$ · t per cycle = 1.5 h · cycles = 4 per week

WHAT YOU WANT

the energy per cycle, in kWh and in joules, and the energy per year

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

What a tumble dryer really costs you in energy

1 A kilowatt-hour is not a unit of power, it is a unit of energy

It is what you get when a kilowatt runs for an hour. Power multiplied by time is always energy, whatever units you use.

$$E = Pt = 2.2 \times 1.5 = 3.3 \text{ kWh per cycle}$$

2 Convert to joules

One kWh is 1000 W running for 3600 s, which is 3.6 million joules.

$$E = 3.3 \times 3\,600\,000 = 11\,880\,000 \text{ J}$$

3 Scale up to a year

$$E = 3.3 \times 4 \times 52 = 686.4 \text{ kWh per year}$$

ANSWER

3.3 kWh (11.9 MJ) per cycle, 686 kWh a year

From one appliance, used four times a week.

⚠ THE TRAP – most students answer 2.2 kWh per cycle

Reading the rating off the label and calling it the energy. The 2.2 kW is a RATE – it tells you nothing until you say for how long. A dryer run for three hours uses twice the energy of one run for ninety minutes, while its rating never changes. Watts describe the appliance; kilowatt-hours describe what you did with it.

DOES IT MAKE SENSE?

A typical UK home uses roughly 2700 kWh of electricity a year. This one appliance accounts for about 25% of that, which is why tumble dryers are the first thing any energy-saving guide tells you to use less.

PROBLEM

Counting the electrons

A current of 0.80 A flows for 2.5 minutes. Find the charge that passes, and the number of electrons it corresponds to. The charge on an electron is 1.60×10^{-19} C.

WHAT YOU KNOW

$$I = 0.80 \text{ A} \cdot t = 2.5 \text{ minutes} \cdot e = 1.60 \times 10^{-19} \text{ C}$$

WHAT YOU WANT

the charge transferred, and the number of electrons

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Counting the electrons

1 Convert the time to seconds first

An amp is defined as a coulomb per SECOND. Minutes have no place inside the equation.

$$t = 2.5 \times 60 = 150 \text{ s}$$

2 Charge is current multiplied by time

$$Q = It = 0.80 \times 150 = 120 \text{ C}$$

3 Divide the total charge by the charge on one electron

$$n = 120 \div (1.60 \times 10^{-19}) = 7.50 \times 10^{20} \text{ electrons}$$

ANSWER

120 C, carried by 7.50×10^{20} electrons

Two and a half minutes of a fairly ordinary current.

⚠ THE TRAP – most students answer $Q = 2.0 \text{ C}$

Leaving the time in minutes. The factor of 60 is the whole mistake, and the answer comes out sixty times too small. Any equation containing an amp, a watt or a volt is written in seconds, because all three are defined per second.

DOES IT MAKE SENSE?

7.50×10^{20} sounds impossibly large until you compare it with a mole, 6.02×10^{23} . It is about a thousandth of a mole of electrons – a sliver of the charge carriers already sitting in the wire, drifting past very slowly indeed.

PROBLEM

Choosing a fuse, and what a fuse is actually protecting

UK mains is 230 V. A kettle is rated at 2800 W and a reading lamp at 60 W. Fuses are available in 3 A, 5 A and 13 A. Choose the correct fuse for each, and say what the fuse is protecting.

WHAT YOU KNOW

$V = 230 \text{ V}$ · $P_{\text{kettle}} = 2800 \text{ W}$ · $P_{\text{lamp}} = 60 \text{ W}$ ·
fuses available = 3, 5, 13 A

WHAT YOU WANT

the right fuse for each appliance, and what the fuse protects

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Choosing a fuse, and what a fuse is actually protecting

1 Find the normal working current of each

$$\text{kettle: } I = P \div V = 2800 \div 230 = 12.2 \text{ A}$$

2 And the lamp

$$\text{lamp: } I = 60 \div 230 = 0.261 \text{ A}$$

3 Pick the smallest fuse ABOVE the working current

Above, so that it does not blow in normal use. The smallest such, so that it reacts as soon as anything goes wrong.

$$\text{kettle } 12.2 \text{ A} \rightarrow 13 \text{ A lamp } 0.261 \text{ A} \rightarrow 3 \text{ A}$$

ANSWER

13 A for the kettle, 3 A for the lamp

And what the fuse protects is the flex, not the appliance.

⚠ THE TRAP – most students answer 13 A for both, on the grounds that a bigger fuse is safer

It is the opposite. A fuse exists to melt before the CABLE overheats, so a 13 A fuse in a lamp flex lets 50 times the normal current flow before it acts – and a thin flex can be glowing long before that. Oversizing a fuse does not make a circuit safer, it makes the fuse ornamental.

DOES IT MAKE SENSE?

Check the kettle from the other direction: a 13 A fuse at 230 V allows up to 2990 W, comfortably above the kettle's 2800 W, while a 5 A fuse would allow only 1150 W and blow the moment it was switched on.

PROBLEM

The force between two charges

A charge of $+2.5 \mu\text{C}$ and a charge of $-4.0 \mu\text{C}$ are held 8.0 cm apart in air. Find the size and nature of the force between them. Take $k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.

WHAT YOU KNOW

$q_1 = +2.5 \mu\text{C} = +2.5 \times 10^{-6} \text{ C}$ · $q_2 = -4.0 \mu\text{C} = -4.0 \times 10^{-6} \text{ C}$ · $r = 8.0 \text{ cm} = 0.080 \text{ m}$

WHAT YOU WANT

the magnitude of the force, and whether it attracts or repels

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The force between two charges

1 Convert everything to SI before touching the equation

Microcoulombs and centimetres both have to go. The centimetre is the dangerous one, because it is squared.

$$q_1 q_2 = (2.5 \times 10^{-6})(4.0 \times 10^{-6}) = 1.0 \times 10^{-11} \text{ C}^2$$

2 Square the separation

$$r^2 = 0.080^2 = 0.0064 \text{ m}^2$$

3 Apply Coulomb's law

Work with the magnitudes and decide attraction or repulsion from the signs afterwards. It is cleaner than carrying a minus through.

$$F = kq_1 q_2 \div r^2 = (8.99 \times 10^9 \times 1.0 \times 10^{-11}) \div 0.0064 = 14.0 \text{ N}$$

4 Opposite signs attract

14.0 N, attractive

ANSWER

14.0 N, attractive

Fourteen newtons between two specks of charge, at the width of a hand.

⚠ THE TRAP – most students answer 1.12 N

Dividing by r instead of r^2 . Coulomb's law is an inverse SQUARE law, like gravity and like light, and forgetting the square here makes the force 12.5 times too small. If an answer to an inverse-square problem does not change by a factor of four when you double the distance, the square is missing.

DOES IT MAKE SENSE?

Apply that very test: at 16 cm the force should be a quarter, 3.50 N. It is. And the sign check is independent of the arithmetic – one charge is positive and one negative, so whatever the size, the force has to pull them together.

PROBLEM

Field and potential are not the same thing

Two parallel plates 2.5 cm apart have a potential difference of 300 V between them. Find the electric field strength between the plates, and the work done in moving a charge of 5.0 nC from one plate to the other.

WHAT YOU KNOW

$V = 300 \text{ V}$ · $d = 2.5 \text{ cm} = 0.025 \text{ m}$ · $Q = 5.0 \text{ nC} = 5.0 \times 10^{-9} \text{ C}$

WHAT YOU WANT

the electric field strength, and the work done moving the charge

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Field and potential are not the same thing

1 Field strength is potential difference PER METRE

That per metre is the whole distinction. Potential is measured at a place; field is how fast potential changes as you move.

$$E = V \div d = 300 \div 0.025 = 12\,000 \text{ V/m}$$

2 Work done is charge multiplied by potential difference

A volt is a joule per coulomb, so coulombs times volts gives joules. The units do the thinking for you.

$$W = QV = (5.0 \times 10^{-9}) \times 300$$

3 Solve

$$W = 1.5 \times 10^{-6} \text{ J} = 1.5 \text{ } \mu\text{J}$$

ANSWER

$E = 12\,000 \text{ V/m}$ and $W = 1.5 \text{ } \mu\text{J}$

A field of twelve thousand volts per metre, from a potential difference of only three hundred volts.

⚠ THE TRAP – most students answer $E = 300 \text{ V/m}$

Quoting the potential difference and attaching the units of field strength. They are different quantities measured in different units: volts against volts per metre. The plates here are only 2.5 cm apart, so the field is 40.0 times the number on the supply. Squeeze the same 300 V into a tenth of the gap and the field is ten times larger again, which is exactly how a spark gap is designed.

DOES IT MAKE SENSE?

Do the work a second way, through the field: $W = QEd = (5.0 \times 10^{-9}) \times 12\,000 \times 0.025 = 1.5 \times 10^{-6} \text{ J}$. The same answer by a different route, which is what you want whenever a problem offers two equations for the same quantity.

PROBLEM

How long a radio wave actually is

A radio station broadcasts on 93.5 MHz. Radio waves travel at 3.00×10^8 m/s. Find the wavelength.

WHAT YOU KNOW

$$f = 93.5 \text{ MHz} \cdot v = 3.00 \times 10^8 \text{ m/s}$$

WHAT YOU WANT

the wavelength of the broadcast

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

How long a radio wave actually is

1 Convert the frequency to hertz first

Mega means a million. The prefix is not decoration, it is a factor of 10^6 .

$$f = 93.5 \times 10^6 \text{ Hz}$$

2 Use the wave equation

True of every wave there is, from ripples to gamma rays.

$$v = f\lambda \rightarrow \lambda = v \div f$$

3 Substitute

$$\lambda = (3.00 \times 10^8) \div (93.5 \times 10^6) = 3.21 \text{ m}$$

ANSWER

3.21 m

Just over three metres, from crest to crest.

⚠ THE TRAP – most students answer $3.21 \times 10^6 \text{ m}$

Leaving the frequency in megahertz. The answer comes out a million times too long – a single wavelength stretching further than the British Isles. Any prefix in a question (M, k, m, μ , n) has to be turned into a power of ten before it goes anywhere near an equation.

DOES IT MAKE SENSE?

FM wavelengths are a few metres, and a quarter-wave aerial is a quarter of that: 0.80 m. That is almost exactly the length of the aerial on a car roof, which is not a coincidence – it is the design.

PROBLEM

Reading a frequency off an oscilloscope

An oscilloscope is set to 2.0 ms per division. One complete cycle of a sound wave spans 4.0 divisions across the screen. Find the period and the frequency.

WHAT YOU KNOW

timebase = 2.0 ms per division · one full cycle =
4.0 divisions

WHAT YOU WANT

the period and the frequency of the wave

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Reading a frequency off an oscilloscope

1 Count the divisions for ONE complete cycle

Crest to crest, or trough to trough. Crest to the next trough is only half a wave, and that is where this question is won or lost.

4.0 divisions per cycle

2 Multiply by the timebase to get the period

$$T = 4.0 \times 2.0 = 8.0 \text{ ms} = 0.0080 \text{ s}$$

3 Frequency is one over the period

Period is seconds per cycle; frequency is cycles per second. They are the same information upside down.

$$f = 1 \div T = 1 \div 0.0080 = 125 \text{ Hz}$$

ANSWER

T = 8.0 ms and f = 125 Hz

A low note, well inside the range of human hearing.

⚠ THE TRAP – most students answer 250 Hz

Measuring from a crest to the next trough and calling it a cycle. That is half a wave, so the period comes out halved and the frequency doubled. On a screen the two look equally like a complete shape, which is why the rule is to measure from one point to the next point that is identical in BOTH height and direction of travel.

DOES IT MAKE SENSE?

125 Hz is a note just below the open B string of a guitar. If the oscilloscope is showing a sound you can hear, the answer has to land between about 20 Hz and 20 000 Hz, and this does.

PROBLEM

Shouting at a cliff

A walker shouts towards a cliff face and hears the echo 1.6 s later. The speed of sound in air is 340 m/s. How far away is the cliff?

WHAT YOU KNOW

$$t = 1.6 \text{ s} \quad v = 340 \text{ m/s}$$

WHAT YOU WANT

the distance to the cliff face

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Shouting at a cliff

1 Work out how far the **SOUND** travelled

Not how far away the cliff is. Those are two different distances, and the question only hands you one of them directly.

$$d = vt = 340 \times 1.6 = 544 \text{ m}$$

2 The sound made the journey twice

Out to the cliff and back to the ear. That is what an echo is.

$$\text{distance to cliff} = 544 \div 2$$

3 Halve it

$$= 272 \text{ m}$$

ANSWER

272 m

The cliff is less than three hundred metres away, though the sound covered twice that.

⚠ THE TRAP – most students answer 544 m

Forgetting the return journey. It is the commonest error in every echo, sonar and radar question there is, and the factor is always exactly two. If a question involves something bouncing back, write $2d = vt$ before you write anything else.

DOES IT MAKE SENSE?

A rough field rule: sound covers about 170 m towards an object for every second of echo delay. $1.6 \text{ s} \times 170 = 272 \text{ m}$, which agrees with the calculation and gives you a way to check it without a calculator.

PROBLEM

Light entering a glass block

A ray of light travelling through air strikes a glass block at 40° to the normal. The refractive index of the glass is 1.50. Find the angle of refraction inside the glass.

WHAT YOU KNOW

angle of incidence = 40° to the normal · n_{glass}
= 1.50 · $n_{\text{air}} = 1.00$

WHAT YOU WANT

the angle of refraction inside the glass

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Light entering a glass block

1 Every angle in optics is measured from the NORMAL

The normal is the line at right angles to the surface. Not from the surface itself – that is the single most expensive habit in this topic.

angle of incidence = 40° from the normal

2 Apply Snell's law

$$n = \sin i \div \sin r \rightarrow \sin r = \sin i \div n$$

3 Substitute

$$\sin r = \sin 40^\circ \div 1.50 = 0.6428 \div 1.50 = 0.4285$$

4 Take the inverse sine

$$r = \sin^{-1} 0.4285 = 25.4^\circ$$

ANSWER

25.4 °

Measured from the normal, inside the glass.

⚠ THE TRAP – most students answer 30.7°

Reading the angle from the surface instead of the normal, so that 40° becomes 50° . The two always add to 90° , which means the wrong value is right there on the diagram looking just as plausible. Draw the normal in, as a dashed line, before you measure anything.

DOES IT MAKE SENSE?

Light entering a denser medium must bend TOWARDS the normal, so the answer has to be smaller than 40° . 25.4° is. Had the answer come out larger, the two sides of Snell's law would have been the wrong way up.

PROBLEM

The angle beyond which light cannot get out

The same glass has a refractive index of 1.50. Find its critical angle, and say what happens to a ray inside the glass that meets the surface at 45° to the normal.

WHAT YOU KNOW

$n_{\text{glass}} = 1.50$ · $n_{\text{air}} = 1.00$ · angle inside the glass = 45° to the normal

WHAT YOU WANT

the critical angle, and the fate of the 45° ray

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The angle beyond which light cannot get out

1 The critical angle is the one that refracts to exactly 90 degrees

At that angle the escaping ray skims along the surface. Any steeper and there is nowhere left for it to go.

$$\sin C = n(\text{outside}) \div n(\text{inside}) = 1.00 \div 1.50$$

2 Solve

$$C = \sin^{-1} 0.6667 = 41.8^\circ$$

3 Compare the ray with the critical angle

$$45^\circ > 41.8^\circ \rightarrow \text{total internal reflection}$$

ANSWER

C = 41.8°, and the 45° ray does not escape at all

It reflects back inside, as completely as from a mirror.

⚠ THE TRAP – most students answer “it still gets out, just bent a long way”

Past the critical angle no light emerges – none, not a dim beam at a steep angle. The refracted ray would have to bend beyond 90°, which would mean going back into the glass it came from. The surface stops behaving like a window and starts behaving like a perfect mirror, which is exactly what makes an optical fibre work.

DOES IT MAKE SENSE?

Run the same calculation for diamond, $n = 2.42$, and the critical angle falls to 24.4°. A much smaller critical angle means light entering a diamond is trapped and bounced around far more before it finds a way out – which is the physics of the sparkle.

PROBLEM

Where a lens puts the image

An object is placed 25 cm from a converging lens of focal length 15 cm. Find the position of the image and its magnification, and describe it.

WHAT YOU KNOW

$u = 25 \text{ cm}$ · $f = 15 \text{ cm}$ · lens = converging

WHAT YOU WANT

the image distance, the magnification, and the nature of the image

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Where a lens puts the image

1 Write the lens equation and rearrange for the unknown

$$1 \div f = 1 \div u + 1 \div v \rightarrow 1 \div v = 1 \div f - 1 \div u$$

2 Substitute

Keep both distances in the same unit and the answer comes out in that unit.

$$1 \div v = 1 \div 15 - 1 \div 25 = 0.02667$$

3 Now INVERT, which is the step that gets forgotten

The equation gives you one over v. It has not given you v.

$$v = 1 \div 0.02667 = 37.5 \text{ cm}$$

4 Magnification is image distance over object distance

$$m = v \div u = 37.5 \div 25 = 1.5$$

ANSWER

$$v = 37.5 \text{ cm}, m = 1.5$$

A real, inverted image, half as big again as the object, on the far side of the lens.

⚠ THE TRAP – most students answer 0.02667 cm

Stopping at $1/v$ and writing it down as the answer. An image distance of 0.02667 cm would put the image inside the glass of the lens itself. The units are the warning: $1/v$ comes out in reciprocal centimetres, and no distance was ever measured in those.

DOES IT MAKE SENSE?

The object sits between f and $2f$, that is between 15 cm and 30 cm. The standard result for that case is an image beyond $2f$, real, inverted and magnified. The calculation puts it at 37.5 cm, which is beyond 30 cm, with a magnification above 1. Every part of the prediction matches.

PROBLEM

The ambulance going past

An ambulance sounds a 850 Hz siren and drives past a stationary observer at 18 m/s. The speed of sound is 340 m/s. Find the frequency heard as it approaches and as it recedes.

WHAT YOU KNOW

$f_{\text{source}} = 850 \text{ Hz}$ · $v_{\text{source}} = 18 \text{ m/s}$ · $v_{\text{sound}} = 340 \text{ m/s}$

WHAT YOU WANT

the frequency heard approaching, and receding

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The ambulance going past

1 The siren still emits 850 waves every second

Nothing about the source changes. What changes is where each wave is emitted from, because the ambulance has moved on between one crest and the next.

$$f(\text{source}) = 850 \text{ Hz throughout}$$

2 Approaching, the crests are squeezed into a shorter distance

Each new crest sets off from closer to you than the last, so the wavelength arriving is shorter and the frequency higher.

$$f = f(\text{source}) \times v \div (v - v(\text{source}))$$

3 Substitute for the approach

$$f = 850 \times 340 \div 322 = 897.5 \text{ Hz}$$

4 Receding, the sign flips

$$f = 850 \times 340 \div 358 = 807.3 \text{ Hz}$$

ANSWER

898 Hz approaching, 807 Hz receding

A drop of 90.2 Hz the instant it passes you.

⚠ THE TRAP – most students answer “the sound travels faster towards you”

It does not. The speed of sound is set by the air, not by whatever made the sound, and it is 340 m/s in every direction regardless of how fast the ambulance is moving. What the motion changes is the WAVELENGTH: the crests pile up ahead of the vehicle and spread out behind it. Same speed, different spacing, different pitch.

DOES IT MAKE SENSE?

The pitch falls from 898 Hz to 807 Hz, a ratio of 1.112. On a musical scale that is close to a whole tone – and a whole tone is about what you hear an ambulance drop as it goes past.

PROBLEM

Why a brighter light does not help

Light of wavelength 400 nm falls on a sodium surface whose work function is 2.3 eV. Find the maximum kinetic energy of the emitted electrons. Take $h = 6.63 \times 10^{-34} \text{ J s}$ and $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$.

WHAT YOU KNOW

$\lambda = 400 \text{ nm} = 400 \times 10^{-9} \text{ m}$ · work function = 2.3 eV · $h = 6.63 \times 10^{-34} \text{ J s}$

WHAT YOU WANT

the maximum kinetic energy of an emitted electron

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Why a brighter light does not help

1 Find the energy of ONE photon

Light arrives in lumps. The whole of the photoelectric effect follows from that one sentence.

$$E = hc \div \lambda = (1.989 \times 10^{-25}) \div (4.00 \times 10^{-7}) = 4.973 \times 10^{-19} \text{ J}$$

2 Convert to electronvolts, because the work function is given in them

$$E = (4.973 \times 10^{-19}) \div (1.60 \times 10^{-19}) = 3.11 \text{ eV}$$

3 The work function is the toll at the door

Some energy is spent simply getting the electron out of the metal. Whatever is left is kinetic energy.

$$\text{KE}(\text{max}) = E - \phi = 3.11 - 2.3 = 0.81 \text{ eV}$$

ANSWER

0.81 eV, which is $1.30 \times 10^{-19} \text{ J}$

And that is the MAXIMUM. Most electrons come out with less.

⚠ THE TRAP – most students answer “turn the light up and the electrons come out faster”

They do not. Brightness is the number of photons per second, not the energy of each one. A brighter lamp of the same colour ejects MORE electrons at exactly the same maximum energy. To give each electron more energy you have to change the colour, not the intensity – and that fact is what killed the wave-only picture of light.

DOES IT MAKE SENSE?

Find the threshold: a 2.3 eV work function needs a photon of at least that energy, so $\lambda = hc \div \phi = 5.40 \times 10^{-7} \text{ m}$, about 540 nm, which is green. So this metal emits under blue and violet light and does nothing at all under red, however bright the red is.

PROBLEM

Why a bright green laser is safer than a faint X-ray

Compare the energy of a photon of green light ($\lambda = 550 \text{ nm}$) with a photon of X-radiation ($\lambda = 0.10 \text{ nm}$). The ionisation energy of a hydrogen atom is 13.6 eV . Which of the two can ionise an atom?

WHAT YOU KNOW

green light = 550 nm · X-ray = 0.10 nm ·
ionisation energy of hydrogen = 13.6 eV

WHAT YOU WANT

the two photon energies, and which can ionise

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Why a bright green laser is safer than a faint X-ray

1 Photon energy is inversely proportional to wavelength

Short wavelength means high energy. That one relationship orders the whole electromagnetic spectrum by how dangerous it is.

$$E = hc \div \lambda$$

2 The green photon

$$E = (1.989 \times 10^{-25}) \div (5.50 \times 10^{-7}) \div (1.60 \times 10^{-19}) = 2.26 \text{ eV}$$

3 The X-ray photon

$$E = (1.989 \times 10^{-25}) \div (1.0 \times 10^{-10}) \div (1.60 \times 10^{-19}) = 12.43 \text{ keV}$$

4 Compare each with 13.6 eV

$$2.26 \text{ eV} < 13.6 \text{ eV} \quad 12430 \text{ eV} > 13.6 \text{ eV}$$

ANSWER

green 2.26 eV, X-ray 12.4 keV - only the X-ray ionises

The X-ray photon carries about 5.5 thousand times the energy of the green one.

⚠ **THE TRAP – most students answer “a powerful laser must be more dangerous than a weak X-ray source”**

Ionisation happens one photon at a time, so what matters is the energy of the individual photon, not how many arrive. A hundred-watt green laser delivers an enormous number of photons, every one of them too feeble to ionise anything – it burns, but it cannot break a molecule apart. A single X-ray photon can.

DOES IT MAKE SENSE?

This is why a reading lamp is harmless and a dental X-ray is measured and shielded. It also explains sunburn: visible light at 2.26 eV does nothing to DNA, while ultraviolet, just beyond the violet end, finally crosses the threshold where chemical bonds start to break.

PROBLEM

A radioactive source, a day later

A radioactive source has an activity of 7200 Bq and a half-life of 6.0 hours. Find its activity 24 hours later.

WHAT YOU KNOW

initial activity = 7200 Bq · half-life = 6.0 hours ·
time elapsed = 24 hours

WHAT YOU WANT

the activity after 24 hours

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A radioactive source, a day later

1 Count how many half-lives have passed

$$n = 24 \div 6.0 = 4 \text{ half-lives}$$

2 Each half-life halves the activity, so they compound

Four halvings is not a quarter and it is not a half four times over. It is a half multiplied by itself four times.

$$A = 7200 \div 2^4 = 7200 \div 16$$

3 Solve

$$A = 450 \text{ Bq}$$

ANSWER

450 Bq

A sixteenth of what it started at, after four half-lives.

⚠ THE TRAP – most students answer 1800 Bq

Dividing by the number of half-lives instead of halving that many times: $7200 \div 4$ rather than $7200 \div 2^4$. Radioactive decay is exponential, not linear. The linear version also predicts that after eight half-lives the activity is negative, which should be enough to retire the method permanently.

DOES IT MAKE SENSE?

After four half-lives the fraction left is $1/16$, which is 6.25% – and 6.25% of 7200 is 450 Bq. Note too that it never reaches zero: halving something repeatedly gets you arbitrarily close and never quite there.

PROBLEM

The mass that goes missing when a nucleus forms

A proton has a mass of 1.007276 u and a neutron 1.008665 u, yet a deuterium nucleus made of one of each has a mass of only 2.013553 u. Find the energy equivalent of the missing mass, in MeV. Take $1 \text{ u} = 1.661 \times 10^{-27} \text{ kg}$ and $c = 3.00 \times 10^8 \text{ m/s}$.

WHAT YOU KNOW

proton = 1.007276 u · neutron = 1.008665 u ·
deuterium nucleus = 2.013553 u · $1 \text{ u} = 1.661 \times 10^{-27} \text{ kg}$

WHAT YOU WANT

the binding energy of the nucleus, in MeV

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

The mass that goes missing when a nucleus forms

1 Find the mass that is unaccounted for

The parts weigh more apart than they do together. That difference is not an error in the measurement; it is the point of the question.

$$\Delta m = (1.007276 + 1.008665) - 2.013553 = 0.002388 \text{ u}$$

2 Convert to kilograms

$$\Delta m = 0.002388 \times (1.661 \times 10^{-27}) = 3.966 \times 10^{-30} \text{ kg}$$

3 Apply $E = mc^2$, and square c

$$E = (3.966 \times 10^{-30}) \times (3.00 \times 10^8)^2 = 3.570 \times 10^{-13} \text{ J}$$

4 Convert to MeV

$$E = (3.570 \times 10^{-13}) \div (1.60 \times 10^{-19}) = 2.23 \text{ MeV}$$

ANSWER

2.23 MeV

The energy you would have to supply to pull the two particles apart again.

⚠ THE TRAP – most students answer $1.19 \times 10^{-21} \text{ J}$

Multiplying by c instead of c^2 . The square is the whole reason nuclear energies are millions of times chemical ones: it turns a factor of 3.00×10^8 into 9.00×10^{16} . Leaving it out makes the answer 3.00×10^8 times too small.

DOES IT MAKE SENSE?

The accepted binding energy of the deuteron is 2.224 MeV. This calculation gives 2.231 MeV from nothing but three masses and a constant – agreement to within a fraction of a percent, which is the strongest possible sign the method is sound.

PROBLEM

An electron with a wavelength

An electron is accelerated from rest through a potential difference of 100 V. Find its de Broglie wavelength. Take the mass of an electron as 9.11×10^{-31} kg and $h = 6.63 \times 10^{-34}$ J s.

WHAT YOU KNOW

$$V = 100 \text{ V} \quad m = 9.11 \times 10^{-31} \text{ kg} \quad e = 1.60 \times 10^{-19} \text{ C} \\ h = 6.63 \times 10^{-34} \text{ J s}$$

WHAT YOU WANT

the de Broglie wavelength of the electron

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

An electron with a wavelength

1 The work done by the field becomes kinetic energy

Charge multiplied by potential difference is energy, and the electron starts from rest, so all of it is kinetic.

$$KE = eV = (1.60 \times 10^{-19}) \times 100 = 1.60 \times 10^{-17} \text{ J}$$

2 Get the speed from the kinetic energy

$$v = \sqrt{(2KE \div m)} = \sqrt{(2 \times (1.60 \times 10^{-17}) \div (9.11 \times 10^{-31}))} = 5.927 \times 10^6 \text{ m/s}$$

3 Momentum, not speed, is what goes into the de Broglie relation

$$p = mv = (9.11 \times 10^{-31}) \times (5.927 \times 10^6) = 5.399 \times 10^{-24} \text{ kg m/s}$$

4 Apply de Broglie

$$\lambda = h \div p = (6.63 \times 10^{-34}) \div (5.399 \times 10^{-24}) = 1.23 \times 10^{-10} \text{ m}$$

ANSWER

$1.23 \times 10^{-10} \text{ m}$, or 0.123 nm

A particle with a wavelength about the width of an atom.

⚠ THE TRAP – most students answer using $\lambda = h \div v$

Dividing by the speed instead of the momentum. The de Broglie relation pairs a wavelength with a MOMENTUM, which is why a cricket ball and an electron at the same speed have utterly different wavelengths – the mass is doing the work. Dropping the mass also leaves the units wrong, and the units are the fastest way to catch it.

DOES IT MAKE SENSE?

Two checks. The wavelength, 0.123 nm, is about the spacing between atoms in a crystal – which is exactly why electrons diffract off crystals and why an electron microscope resolves what light never can. And the speed came out at 1.98% of the speed of light, comfortably low enough that ignoring relativity was justified.

ANSWER KEY

Numbers only. If yours does not match, the method is on the solution page – and so is the reason you got the one you got.

1	31.3 m/s	14	$v = 1.20 \text{ m/s}$, and 2.16 J lost
2	8000 kg/m ³	15	24 500 N without the airbag, 3920 N with it
3	10.0 kN at 36.9° east of north	16	13.1 m/s
4	5.0 m/s	17	138 kPa
5	75.0 m	18	it floats, with 56.7% of it submerged
6	3.03 s	19	6750 N, enough to lift 688 kg
7	$t = 3.03 \text{ s}$ and range = 36.3 m	20	16.7 N
8	755 N	21	4.8 m/s, at a pressure 10 800 Pa LOWER
9	it slides, at 1.48 m/s ²	22	514 kJ, taking 184 s
10	10.2 N	23	83.5 kJ
11	779 J	24	42.5 °C
12	7.92 m/s	25	3.75 bar absolute
13	368 W		

ANSWER KEY

Numbers only. If yours does not match, the method is on the solution page – and so is the reason you got the one you got.

26	2.17 cm ³	39	3.21 m
27	39.6% efficient, rejecting 1450 MW	40	T = 8.0 ms and f = 125 Hz
28	73.3 W, which is 6.33 MJ or 1.76 kWh per day	41	272 m
29	12.0 Ω, dissipating 0.750 W	42	25.4 °
30	21.8 mA, with 4.80 V and 7.20 V across them	43	C = 41.8°, and the 45° ray does not escape at all
31	132 Ω, drawing 90.9 mA	44	v = 37.5 cm, m = 1.5
32	2.88 V	45	898 Hz approaching, 807 Hz receding
33	6.86 W	46	0.81 eV, which is 1.30×10^{-19} J
34	3.3 kWh (11.9 MJ) per cycle, 686 kWh a year	47	green 2.26 eV, X-ray 12.4 keV – only the X-ray ionises
35	120 C, carried by 7.50×10^{20} electrons	48	450 Bq
36	13 A for the kettle, 3 A for the lamp	49	2.23 MeV
37	14.0 N, attractive	50	1.23×10^{-10} m, or 0.123 nm
38	E = 12 000 V/m and W = 1.5 μJ		

EVERY TRAP, IN ONE PLACE

Read this page the night before. It is the shortest revision in the book.

FUNDAMENTALS

- 1 Dividing by 3.6, which converts km/h and not mph
- 2 Changing the unit label without changing the number
- 3 Adding the magnitudes of two perpendicular forces
- 4 Dividing distance by time at one point, on a line that misses the origin

MECHANICS

- 5 Putting a speed in mph straight into SUVAT
- 6 Dividing the height by g , as though g were a speed
- 7 Thinking horizontal speed keeps a projectile up for longer
- 8 Writing $N = ma$ and forgetting the weight
- 9 Using $N = mg$ on a slope
- 10 Treating μN as the friction force rather than its maximum
- 11 Forgetting $\cos \theta$ when the pull is at an angle
- 12 Calling the question unanswerable because the mass is missing
- 13 Quoting the work done and labelling it power
- 14 Conserving kinetic energy in a collision where things stick
- 15 Believing the airbag reduces the momentum change
- 16 Thinking a heavier car has to corner more slowly

FLUIDS

- 17 Giving the gauge pressure when the total was asked for
- 18 Thinking a bigger block of the same material sinks
- 19 Believing a hydraulic jack multiplies energy
- 20 Using the object's density in the upthrust instead of the fluid's
- 21 Expecting higher pressure where the pipe is narrower

EVERY TRAP, IN ONE PLACE

Read this page the night before. It is the shortest revision in the book.

THERMODYNAMICS

- 22 Using the final temperature as ΔT
- 23 Expecting the temperature to rise while ice melts
- 24 Averaging two temperatures instead of balancing the energy
- 25 Taking a gas-law ratio in degrees Celsius
- 26 Leaving atmospheric pressure out of Boyle's law
- 27 Calling all of a power station's rejected heat bad engineering
- 28 Swapping a rate for a total, watts for joules

ELECTRICITY

- 29 Dividing I by V instead of V by I
- 30 Splitting the supply voltage equally between unequal resistors
- 31 Adding resistances that are in parallel
- 32 Putting the wrong resistor on top in a potential divider
- 33 Reaching for $P = VI$ and using the supply voltage
- 34 Reading the power rating off the label and calling it the energy
- 35 Leaving the time in minutes
- 36 Fitting a bigger fuse on the grounds that it must be safer
- 37 Dividing by r instead of r^2 in an inverse-square law
- 38 Quoting the potential difference as the field strength

EVERY TRAP, IN ONE PLACE

Read this page the night before. It is the shortest revision in the book.

WAVES & OPTICS

- 39 Leaving the frequency in megahertz
- 40 Measuring half a wave and calling it a cycle
- 41 Forgetting that an echo makes the journey twice
- 42 Measuring the angle from the surface instead of the normal
- 43 Expecting light to escape beyond the critical angle
- 44 Stopping at $1/v$ and writing it down as the distance
- 45 Thinking sound travels faster towards you

MODERN PHYSICS

- 46 Expecting a brighter light to give each electron more energy
- 47 Judging danger by the brightness rather than the photon energy
- 48 Dividing by the number of half-lives instead of halving that often
- 49 Multiplying by c instead of c^2
- 50 Dividing h by the speed instead of by the momentum

WHERE TO GO NEXT

This book was built to sit beside the others, not to replace them.

Physics Study Lab – the Kit

Seven parts and thirty-eight illustrated chapters: the theory behind every problem in this book. The footer of each solution tells you the page. If a step did not click, start there.

The Words of Physics

104 terms, each with its own drawing and its own why. Most of what looks like a physics problem is a vocabulary problem wearing a disguise – see the command words page.

Beyond the Syllabus

Quantum physics, capacitance and RC circuits, gravitational fields, particle physics and cosmology, derived rather than merely defined. The next step up from the hardest problems here.

The Final Book of Physics

127 pages, twelve parts, from Newton's laws to dark energy, plus a master reference of every formula, constant, unit and symbol. The one to keep when the exams are over.

Understanding physics and being able to use it are two different skills.

You have now practised the second one fifty times.